

Assignment-02

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Sec : 21

CSE422

Ans. to the ques. no. 1

i)

<u>Field</u>	<u>Gender</u>	<u>Level</u>	
		<u>Senior</u>	<u>Junior</u>
Cybersecurity	Female	0.12	0
	Male	0.09	0.02
Data science	Female	0	0.15
	Male	0.07	0.18
Cloud computing	Female	0.17	0.04
	Male	0.13	0.03
Total			1.00

$$ii) P(\neg \text{Data science} | \text{Junior}) = \frac{P(\neg \text{Data science} \cap \text{Junior})}{P(\text{Junior})}$$

$$= \frac{0.02 + 0.04 + 0.03}{0 + 0.02 + 0.15 + 0.18 + 0.04 + 0.03}$$

$$= \frac{0.09}{0.42}$$

$$= 0.214$$

iii) Given,

$$F \perp L | G = \text{Female}$$

We need to find out if Level matter in predicting field when we know that the gender is female.

$$\therefore P(F | L, G) = P(F | G)$$

$$P(G = \text{Female}) = 0 + 0.12 + 0 + 0.15 + 0.17 + 0.04$$

$$= 0.48$$

For different fields $P(F|G)$ —

$$P(F = \text{Cybersecurity} | G = \text{Female}) = \frac{0.12}{0.48} = 0.25$$

$$P(L = \text{Data science} | \text{Female}) = \frac{0.15}{0.48} = 0.312$$

$$P(L = \text{Cloud computing} | \text{Female}) = \frac{0.17 + 0.04}{0.48} = 0.437$$

Now for $P(L|G)$ —

$$P(L = \text{Senior} | G = \text{Female}) = 0.12 + 0 + 0.17 = 0.29$$

$$P(L = \text{Junior} | G = \text{Female}) = 0 + 0.15 + 0.04 = 0.19$$

For different field —

Senior

$$P(F = \text{Cybersecurity} | L, G) = \frac{0.12}{0.29} = 0.414$$

$$P(L = \text{Data science} | L, G) = \frac{0}{0.29} = 0$$

$$P(L = \text{Cloud computing} | L, G) = \frac{0.17}{0.29} = 0.586$$

Junior

$$P(\text{Cybersecurity} | L, G) = \frac{0}{0.19} = 0$$

$$P(\text{Data science} | L, G) = \frac{0.15}{0.19} = 0.789$$

$$P(\text{Cloud computing} | L, G) = \frac{0.04}{0.19} = 0.211$$

Here, none of the values matches. That is, even if one value does not match, the condition is not met.

That is,

F & L are not conditionally independent given $G = \text{Female}$.

iv) The condition is —

$$\text{Expected outcome} = 25 \times P(\text{Data science} | \text{Senior})$$

Now,

$$P(\text{Senior}) = 0.12 + 0.09 + 0 + 0.07 + 0.17 + 0.13$$

$$= 0.58$$

$$P(\text{Data science} \cap \text{Senior}) = 0.12 + 0.09 + 0.17 + 0.13$$

$$= 0.51$$

$$\therefore P(\text{Data science} | \text{Senior}) = \frac{P(\text{Data science} \cap \text{Senior})}{P(\text{Senior})}$$

$$= \frac{0.51}{0.58}$$

$$= 0.879$$

$$\therefore E = 25 \times 0.879$$

$$= 21.97 \approx 22 \text{ students}$$

Ans. to the ques. no. 2

Given,

$$P(A|B, C) = P(B|A, C) \cdot 0 + 0 + 0 + 1 \cdot 0 = 0$$

$$\text{or, } \frac{P(A \cap B \cap C)}{P(B \cap C)} = \frac{P(A \cap B \cap C)}{P(A \cap C)}$$

$$\text{or, } \frac{1}{P(B \cap C)} = \frac{1}{P(A \cap C)} \quad [\text{Removing } P(A \cap B \cap C) \text{ from both side}]$$

$$\text{or, } P(A \cap C) = P(B \cap C)$$

$$\text{or, } \frac{P(A \cap C)}{P(C)} = \frac{P(B \cap C)}{P(C)} \quad [\text{Dividing both side by } P(C)]$$

$$\text{or, } P(A|C) = P(B|C)$$

Ans. to the ques. no. 3

$$P(P_{os} | D) = 0.91$$

$$P(\neg P_{os} | D) = 0.09$$

$$P(P_{os} | \neg D) = 0.12$$

$$P(\neg P_{os} | \neg D) = 0.88$$

$$P(D) = 0.18$$

$$P(\neg D) = 0.82$$

$$a) P(\neg D | P_{os}) = \frac{P(P_{os} | \neg D) \cdot P(\neg D)}{P(P_{os})}$$

Now,

$$P(P_{os}) = P(P_{os} | D) \cdot P(D) + P(P_{os} | \neg D) \cdot P(\neg D)$$

$$= 0.91 \times 0.18 + 0.12 \times 0.82$$

$$= 0.26$$

That is,

$$P(\neg D | P_{os}) = \frac{0.12 \times 0.82}{0.26} = 0.37$$

$$\begin{aligned}
 b) \quad P(P_{os}, \neg P_{os}, \neg P_{os} | D) &= P(P_{os} | D) \cdot P(\neg P_{os} | D) \cdot P(\neg P_{os} | D) \\
 &= 0.91 \times 0.09 \times 0.09 \\
 &= 0.0073
 \end{aligned}$$

$$\begin{aligned}
 \therefore P(D | P_{os}, \neg P_{os}, \neg P_{os}) &= P(P_{os}, \neg P_{os}, \neg P_{os}) \cdot P(D) \\
 &= 0.0073 \times 0.18 \\
 &= 0.0013
 \end{aligned}$$

$$\begin{aligned}
 P(P_{os}, \neg P_{os}, \neg P_{os} | \neg D) &= P(P_{os} | \neg D) \cdot P(\neg P_{os} | \neg D) \cdot P(\neg P_{os} | \neg D) \\
 &= 0.12 \times 0.88 \times 0.88 \\
 &= 0.093
 \end{aligned}$$

$$\begin{aligned}
 \therefore P(\neg D | P_{os}, \neg P_{os}, \neg P_{os}) &= P(P_{os}, \neg P_{os}, \neg P_{os} | \neg D) P(\neg D) \\
 &= 0.093 \times 0.82 \\
 &= 0.076
 \end{aligned}$$

Since

$$0.076 > 0.0013$$

$\therefore$ Mr. Rahman is not infected.

Ans. to the ques. no. 4

a) In linear regression, w and w_0 are the parameters (coefficient) that define the regression model.

$$y = wx + w_0$$

Here w represent the slope of the line. It tells us how much the predicted output y changes when the input x is increased or decreased. It determines the direction and steepness of the line.

On the other hand, w_0 represents the bias as well as the value of y when $x=0$. This shifts the regression line upward or downward.

b) The first datapoints are -

$$x = 1000, y = 50$$

Given,

$$\omega_1 = 0.03$$

$$b = 0$$

$$\alpha = 0.000001$$

First iteration

$$\begin{aligned}\frac{\partial L}{\partial \omega} &= -2x(y - \hat{y}_1) \\ &= -2 \times [1000 \times (50 - 30)] \\ &= -2(20000) \\ &= -40000\end{aligned}$$

$$\begin{aligned}\hat{y}_1 &= \omega_1 x + b \\ &= 0.03 \times 1000 \\ &= 30\end{aligned}$$

$$\begin{aligned}\frac{\partial L}{\partial b} &= -2(y - \hat{y}) \\ &= -2(20) \\ &= -40\end{aligned}$$

Update -

$$\omega = \omega_{old} - \alpha \frac{\partial L}{\partial \omega} = 0.03 - 0.000001(-40000) = 0.07$$

$$b = b_{old} - \alpha \frac{\partial L}{\partial b} = 0 - 0.000001(-40) = 4 \times 10^{-5}$$

Iteration 2

$$\begin{aligned}\frac{\partial L}{\partial w} &= -2[1000(50-70)] \\ &= 40000\end{aligned}$$

$$\begin{aligned}\frac{\partial L}{\partial b} &= -2(50-70) \\ &= 40\end{aligned}$$

Update -

$$\begin{aligned}w &= 0.07 - 0.000001(40000) \\ &= 0.03\end{aligned}$$

$$\begin{aligned}b &= 4 \times 10^{-5} - 0.000001(40) \\ &= 0\end{aligned}$$

$$\begin{aligned}\hat{y} &= w_1x + b \\ &= 0.03 \times 1000 + 0 \\ &= 30\end{aligned}$$

That is, after 2 iterations, the weights turn back to its original.

c) After the 2nd iteration,

$$\omega_1 = 0.03$$

$$b = 0$$

$$\begin{aligned}\therefore \hat{y}(2100) &= \omega_1 x + b \\ &= 0.03(2100) + 0 \\ &= 63 \text{ lakh taka}\end{aligned}$$

Given,

2500 sq. feet requires 125 lakh taka

$$\therefore 1 \quad " \quad " \quad \frac{125}{2500}$$

$$= 0.05 \text{ lakh taka}$$

That is,

$$\text{Expected price based on dataset trend} = 2100 \times 0.05$$

$$= 105 \text{ lakh taka}$$

Thus it seems that the model underpredicts, it has not converged yet

d) When loss is minimal, we are at the bottom of the loss curve. The slope (derivative) is zero at minimum. Gradients being large often refers that the model is still far from reaching the optimum, indicating that the prediction will be poor. Gradients being minimum or zero indicates that no further improvement is required, meaning the optimum is reached and predictions are accurate. It also indicates that we've found the best weights. Thus, the derivative of the loss function with respect to each weight should be close to zero for achieving better model prediction.

e) The effects of increasing learning rate on gradient descent are as such -

- i) Training becomes unstable and model never learns properly.
- ii) The model overshoots the minimum each time.
- iii) The loss may increase rather than decrease.
- iv) The update of w , and b gets larger value everytime.
- v) Instead of converging, the weights oscillate or diverge.

Ans. to the ques. no. 5

a) Given,

$$\omega_1 = 0.3$$

$$\omega_2 = -0.2$$

$$b = 0$$

$$x_1 = 2$$

$$x_2 = 1$$

We know,

$$z = \omega_1 x_1 + \omega_2 x_2 + b$$

$$= 0.3(2) + (-0.2)\left(\frac{1}{2}\right) + 0$$

$$= 0.4$$

$$\hat{y} = \sigma(z)$$

$$= \frac{1}{1 + e^{-0.4}}$$

$$= 0.598$$

b) Given,

$$y = 1$$

$$\alpha = 0.1$$

We know,

$$\begin{aligned}\frac{\partial L}{\partial \omega_1} &= x_1 (\hat{y} - y) \\ &= 2(0.598 - 1) \\ &= -0.804\end{aligned}$$

Update -

$$\begin{aligned}\omega_1 &= \omega_{1(\text{old})} - \alpha \frac{\partial L}{\partial \omega_1} \\ &= 0.3 - 0.1(-0.804) \\ &= 0.3804\end{aligned}$$

c) The effects of using a decaying learning rate while training a logistic learning model are as follows -

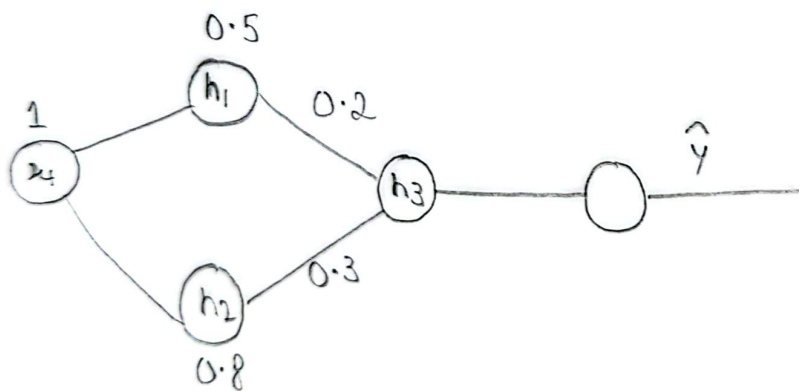
- i) Small α makes sure that model takes tiny step and converges without overshooting.
- ii) Large α allows the model to take big steps and explore

the loss surface more quickly at first.

iii) The model learns fast at first and then fine tunes. This leads to better final accuracy and more stable convergence.

iv) It avoids both the oscillation problem with large α as well as extreme slowness with small α .

Ans. to the ques. no. 6



a) Given,

$$x_1 = 1$$

$$b = 0$$

$$h_1 = 0.5$$

$$h_2 = 0.8$$

$$\omega_1 = 0.2$$

$$\omega_2 = 0.3$$

Now, for hidden layer-2,

$$z = \omega_1 h_1 + \omega_2 h_2 + b$$

$$= 0.2(0.5) + 0.3(0.8) + 0$$

$$= 0.34$$

b) Activation output for hidden layer-02-

$$h_3 = \sigma(z) = \frac{1}{1 + e^{-0.34}}$$

$$= 0.584$$

c) In a neural network, there can be multiple number of hidden layer regardless of the number of input features. Here,

there is only one input feature, while hidden layer-1 has

2 neurons. These neurons have their very own set of weight and bias. They also independently compute -

$$h_i = \sigma(\omega_i x_1 + b_i)$$

Here, each neuron learns different linear transformation of the same input. This allows it to capture different features or patterns, even from a single value. Here, the number of outputs from a layer doesn't come from the number of inputs, but rather the number of neurons in that layer.